Same Legal Facts, Different Verdicts: Evaluating Cross-Lingual and Dialectal Robustness in LLM Legal Reasoning

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Abstract

Large language models (LLMs) are increasingly evaluated on legal reasoning tasks, but most benchmarks remain English-centered and test each legal problem in only one linguistic form. We examine whether LLMs preserve final-answer accuracy when the same legal question is expressed in English, Modern Standard Arabic, Simple Arabic, and Egyptian Arabic. We construct a controlled benchmark of 200 unique MBE-style multiple-choice problems represented in four semantically aligned versions, yielding 800 linguistic instances. The legal facts, answer choices, legal issue, and correct answer are preserved across versions through human translation and legal and linguistic validation. We evaluate GPT-5, Gemini 2.5 Pro, Grok, and Qwen using standardized language-matched prompts and a shared, human-verified answer key. All models achieve their highest accuracy in English, but their Arabic performance varies substantially. Qwen scores 97.0% in English and 79.0% in each Arabic condition, while GPT-5 exhibits the greatest cross-lingual stability, scoring 90.0% in English and 80.0–82.0% across the Arabic conditions. Gemini 2.5 Pro declines from 80.5% in English to 17.5–21.0% in Arabic, while Grok declines from 45.0% to 16.5–26.5%. These results demonstrate substantial cross-lingual final-answer gaps within this translated MBE-style setting. They do not establish robustness across legal systems or equivalence in the models' underlying reasoning processes. Our findings show that English-language accuracy alone is insufficient to assess model reliability in Arabic legal question answering.

1 Introduction

Large language models (LLMs) are increasingly evaluated on legal reasoning tasks, including bar-exam-style questions, statutory interpretation, rule application, and legal judgment prediction. Prior studies have reported strong performance on English legal benchmarks, including LegalBench and simulated bar examinations (Guha et al., 2023; Katz et al., 2024; Alsagheer et al., 2025). However, most legal evaluations remain English-centered and present each legal problem in only one linguistic form. Consequently, high English accuracy does not establish that a model will preserve its performance when the same problem is expressed in another language, register, or dialect.

This limitation is especially important for Arabic-speaking users. Arabic is not a single uniform linguistic condition: users may interact with LLMs in Modern Standard Arabic (MSA), simplified registers, regional dialects, or mixed forms. Multilingual benchmarks have shown that strong English performance does not necessarily imply comparable cross-lingual generalization (Hu et al., 2020; Liang et al., 2020; Bandarkar et al., 2024). Arabic benchmarks such as LAraBench and ArabicMMLU further demonstrate the importance of language- and variety-sensitive evaluation (Abdelali et al., 2024; Koto et al., 2024). In the legal domain, ALARB evaluates Arabic legal argument reasoning (Abu Shairah et al., 2025), but existing work does not directly test whether LLMs preserve their answers when the same legal problem is presented in semantically aligned English and Arabic forms.

Legal multiple-choice questions provide a demanding setting for this evaluation because the correct answer may depend on precise interactions among facts, rules, exceptions, procedural posture, and closely related answer choices. Even a small semantic change can legitimately alter the answer. An uncontrolled comparison may therefore conflate model sensitivity with translation differences. To address this problem, we construct a paired benchmark in which every item preserves the legal facts, legal issue, answer choices, answer order, and gold answer across four linguistic conditions: English, Original Arabic/Modern Standard Arabic (OA/MSA), Simple Arabic (SA), and Egyptian

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Arabic (EA).

The benchmark contains 200 unique MBE-style legal problems, each represented in four aligned versions, yielding 800 linguistic instances: 200 English and 600 Arabic instances. These are not 800 independent legal problems. Reusing the same problems enables controlled within-item comparisons in which the legal content remains fixed while the linguistic form changes.

The Arabic versions were translated by humans and underwent legal and linguistic validation. Two bilingual lawyers reviewed every version for the preservation of the facts, the applicable legal issue, the rule's application, the answer-choice meanings, and the correct answer. A Ph.D.-level Arabic-language reviewer separately assessed fluency, clarity, register appropriateness, and dialectal naturalness. Items identified as ambiguous or misaligned were revised and re-reviewed before evaluation.

Simple Arabic was included as a readability-oriented condition to examine whether reducing linguistic complexity improves model performance without changing the legal task. Egyptian Arabic was selected because it is widely understood and because the research team had the expertise needed to validate it rigorously. We do not assume that the observed patterns generalize to Gulf, Levantine, Maghrebi, or other Arabic varieties; evaluating additional dialects and mixed MSA–dialect forms remains important future work.

We frame the measured property as *cross-lingual robustness* because the legal facts, issue, answer choices, and gold answer are held constant while only the linguistic form changes. Under this controlled transformation, a robust model should preserve its final-answer performance. The resulting disparities also raise fairness and access concerns for Arabic-speaking users, but fairness itself is not directly measured in this study. Moreover, exact-match accuracy evaluates final-answer correctness rather than the faithfulness, legal validity, or cross-language equivalence of the models' underlying reasoning processes.

We investigate the following research questions:

RQ1: How accurately do LLMs answer the same legal questions in English, OA/MSA, Simple Arabic, and Egyptian Arabic?

RQ2: How large are the English–Arabic performance gaps when the legal content and answer key are held constant?

RQ3: Which evaluated models are most robust to changes in Arabic register and dialect?

RQ4: Does simplifying Arabic improve final-answer accuracy, or can it introduce additional instability?

To answer these questions, we evaluate GPT-5 (OpenAI, 2025), Gemini 2.5 Pro (Google DeepMind, 2025), Grok 3 (xAI, 2025), and Qwen 2 (Yang et al., 2024) using standardized, language-matched prompts and a shared, human-verified answer key. We report language-specific accuracy, average Arabic accuracy, English–Arabic performance gaps, maximum cross-language variation, and Wilson 95% confidence intervals. All four models achieve their highest accuracy in English, but their Arabic performance differs substantially. Qwen achieves the highest English accuracy at 97.0% but scores 79.0% in each Arabic condition. GPT-5 exhibits the greatest cross-lingual stability, scoring 90.0% in English and 80.0–82.0% across Arabic conditions. Gemini 2.5 Pro declines from 80.5% in English to 17.5–21.0% across Arabic conditions, while Grok declines from 45.0% to 16.5–26.5%.

This work makes three principal contributions:

1. We introduce a controlled paired evaluation design containing 200 unique MBE-style legal problems and 800 semantically aligned English and Arabic instances.

2. We evaluate four widely used LLM interfaces using language-specific accuracy and complementary measures of cross-lingual final-answer robustness.

3. We provide empirical evidence that English accuracy alone is insufficient for assessing model reliability on semantically aligned Arabic legal questions.

Our conclusions concern cross-lingual final-answer robustness on translated MBE-style multiple-choice questions. The study does not compare different legal systems, evaluate Arabic legal doctrine, assess open-ended legal advice or professional judgment, or determine whether models use equivalent reasoning processes across languages. The findings should therefore be interpreted within the evaluated task, source domain, linguistic conditions, prompts, and model interfaces.

2 Related Work

Legal reasoning benchmarks for LLMs. Recent work has developed benchmarks for evaluating the legal capabilities of large language models (LLMs). LegalBench provides a collaboratively constructed collection of legal reasoning tasks designed by legal professionals and evaluated across multiple LLMs (Guha et al., 2023). Other studies assess LLM performance on professional legal examinations, including the performance of GPT-4 on the Uniform Bar Examination (Katz et al., 2024). These studies demonstrate substantial progress on English-language legal tasks, but they primarily measure legal competence in English. They do not test whether a model preserves its answer when the same legal problem is presented across English and multiple Arabic language varieties.

Multilingual and Arabic LLM evaluation. Multilingual benchmarks such as XTREME, XGLUE, and Belebele demonstrate that strong English-language performance does not necessarily imply comparable cross-lingual generalization (Hu et al., 2020; Liang et al., 2020; Bandarkar et al., 2024). Arabic evaluation introduces additional challenges because Arabic is used in Modern Standard Arabic, regional dialects, code-switched forms, and registers that vary in linguistic complexity. Arabic benchmarks such as LAraBench and ArabicMMLU evaluate LLM performance across a broad range of Arabic-language tasks and domains (Abdelali et al., 2024; Koto et al., 2024). However, these benchmarks are general-purpose evaluations and are not designed as controlled, within-item tests in which the legal facts, answer choices, and correct answer are preserved across English and multiple Arabic versions.

Arabic-centric language models. In parallel with benchmark development, recent technical reports have introduced language models specifically designed to strengthen Arabic capabilities. ALLaM is a family of Arabic–English language models that uses bilingual pretraining, vocabulary expansion, parallel data, and preference alignment to improve Arabic performance while retaining English capabilities (Bari et al., 2024). Fanar is an Arabic-centric multimodal generative AI platform whose language-model components are trained on large-scale Arabic, English, and code corpora (Fanar Team, 2025). ALLaM and Fanar are models rather than evaluation benchmarks and therefore represent complementary work to Arabic benchmark development. Our experiments evaluate four general-purpose model interfaces rather than Arabic-centric models. Evaluating ALLaM, Fanar, and other Arabic-focused or fixed open-weight models under the same controlled legal protocol is an important direction for future work. Such comparisons could help determine whether Arabic-focused pretraining and alignment reduce the cross-lingual final-answer gaps observed in the present study.

Work	Leg.	Ar.	Align.	Var.
LegalBench (Guha et al., 2023)	Y	N	N	N
Bar Exam (Katz et al., 2024)	Y	N	N	N
LAraBench (Abdelali et al., 2024)	N	Y	N	Y
ArabicMMLU (Koto et al., 2024)	N	Y	N	N
MultiLegalPile (Niklaus et al., 2024)	Y	Part.	N	N
ALARB (Abu Shairah et al., 2025)	Y	Y	N	N
Ours	**Y**	**Y**	**Y**	**Y**

Table 1: Comparison of related legal and Arabic benchmarks. Leg.: legal focus; Ar.: Arabic coverage; Align.: aligned questions; Var.: Arabic varieties.

Multilingual legal resources. Research has also expanded the availability of multilingual legal NLP resources. MultiLegalPile provides a large multilingual legal corpus spanning multiple languages and jurisdictions (Niklaus et al., 2024). Within Arabic legal NLP, ALARB introduces an Arabic legal argument-reasoning benchmark derived from Saudi commercial court cases and includes tasks involving verdict prediction, legal-reasoning completion, and regulation identification (Abu Shairah et al., 2025). These resources advance multilingual and Arabic legal AI, but they address objectives different from ours. MultiLegalPile primarily supports multilingual legal language modeling, while ALARB evaluates legal reasoning within an Arabic legal setting. Neither is designed to isolate the effect of linguistic form by presenting the same legal problem across aligned English, Modern Standard Arabic, Simple Arabic, and Egyptian Arabic versions.

Table 1 shows that prior resources do not jointly evaluate legal reasoning, Arabic variation, and aligned versions of the same problem. Our paired design holds the legal content and gold answer constant to measure cross-lingual final-answer robustness, not robustness across legal systems or reasoning-process equivalence.

3 Methodology

This section describes the construction of our controlled multilingual legal reasoning benchmark and

the evaluation procedure used to test whether LLMs preserve legal reasoning performance across English, formal Arabic, simplified Arabic, and Egyptian Arabic. Our goal is to isolate the effect of language, register, and dialect while keeping the underlying legal task fixed. Figure 1 summarizes the full evaluation pipeline.

3.1 Dataset Source and Benchmark Design

The English source questions were selected from protected National Conference of Bar Examiners (NCBE) Multistate Bar Examination (MBE) preparation materials. MBE-style questions are suitable for this study because each item contains a legal fact pattern, four answer choices, and one designated correct answer. Prior work has similarly used bar examinations and legal benchmarks to evaluate LLM legal reasoning (Katz et al., 2024; Guha et al., 2023), although such evaluations remain largely English-centered.

We selected 200 questions from a larger pool. We retained items with a complete fact pattern, four well-formed answer choices, and a clearly specified correct answer. We excluded items with incomplete or malformed content, ambiguous answers, English-specific wording, or legal distinctions that could not be reliably preserved in Arabic. The selected questions were approximately balanced across the principal MBE subject areas, with greater representation of Civil Procedure.

Each question was represented in English, Modern Standard Arabic (MSA), Simple Arabic, and Egyptian Arabic, producing 800 aligned instances: 200 English and 600 Arabic. These instances represent 200 unique legal problems, not 800 independent problems. Across the four versions, the legal facts, legal issue, answer choices, answer order, and gold answer were held constant through the legal and linguistic validation process described below. This paired design isolates changes associated with language, register, and dialect while keeping the underlying legal task fixed.

Because the source questions and official explanations are protected NCBE materials, we cannot reproduce, publicly redistribute, or release the complete benchmark, including the original English questions, their Arabic translations, answer choices, and official explanations. We instead report aggregate results, evaluation procedures, prompt templates, scoring rules, and non-reconstructive synthetic examples. Further details about data availability and reproducibility are provided in Section 6.

3.2 Language Conditions

Each legal problem appears in four aligned language conditions:

- **English (EN):** the original source version.
- **Modern Standard Arabic (MSA/OA):** a formal Arabic version preserving the structure and legal meaning of the English item.
- **Simple Arabic (SA):** a readability-oriented version using simpler wording and sentence structure while preserving the same legal task.
- **Egyptian Arabic (EA):** a dialectal version written in natural Egyptian Arabic while preserving the same legal content.

These conditions test whether model performance changes when the linguistic form varies while the underlying legal problem and correct answer remain fixed.

3.3 Arabic Translation and Validation

All Arabic versions were human-translated. Two bilingual lawyers reviewed every MSA, Simple Arabic, and Egyptian Arabic version to verify preservation of the legal facts, issue, applicable rule, answer-choice meanings, answer order, and gold answer. A Ph.D.-level Arabic-language reviewer separately evaluated each version for fluency, clarity, register appropriateness, and dialectal naturalness.

When a reviewer identified ambiguity, unnatural wording, mistranslation, or a possible change in legal meaning, the translation and review team discussed and revised the item. Every revised item was then re-reviewed for both legal and linguistic equivalence before inclusion in the benchmark.

3.4 Evaluated Models and Prompting

We evaluated four general-purpose LLMs: GPT-5, Gemini 2.5 Pro, Grok 3, and Qwen 2. These models were selected because they represent widely used commercial or publicly accessible LLM families with multilingual capabilities. They were accessed via the official user-facing interfaces for ChatGPT, Gemini, Grok, and Qwen Studio, respectively. We recorded the displayed model identifier and access date when available; otherwise, we recorded the interface's default model.

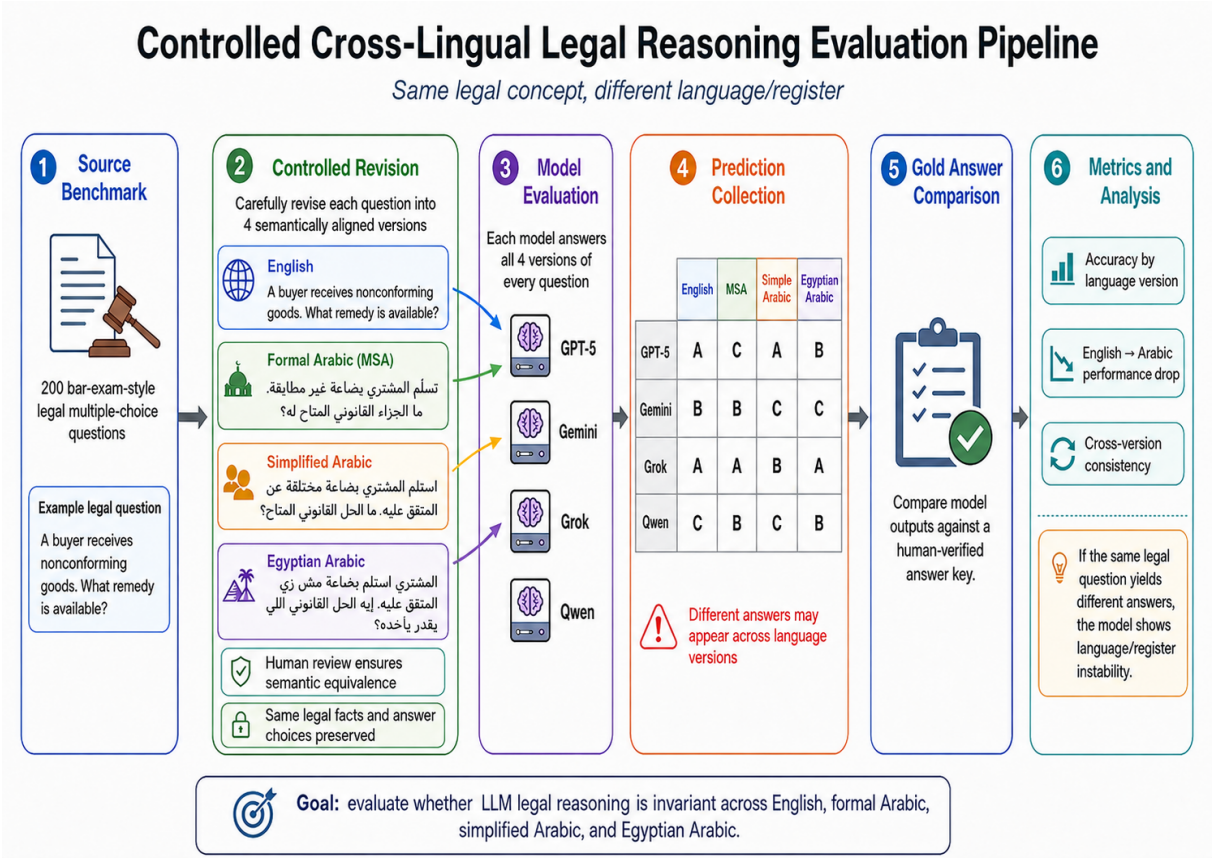


Figure 1: Controlled evaluation pipeline for legal reasoning across English and three Arabic language varieties.

Each model was evaluated on the same 800 aligned instances: 200 English, 200 Original Arabic/MSA, 200 Simple Arabic, and 200 Egyptian Arabic. We accessed all models in May 2026 and performed one run per model–item–language combination, yielding 800 responses per model and 3,200 responses overall. No repeated sampling or majority voting was used.

We used each interface's default settings. Web browsing, retrieval, external documents, file uploads, memory, and legal-assistance tools were not enabled. The results therefore represent interface-level behavior during the evaluation period rather than deterministic fixed-checkpoint performance.

Before the full evaluation, we piloted several instruction variants for output compliance and then fixed one concise template:

> Read the following legal multiple-choice question and select the single best answer. Return only the answer letter: A, B, C, or D.

For Arabic conditions, the instruction was translated into the corresponding Arabic variety. Each prompt contained one complete legal fact pattern and four choices labeled A–D. The question language and instruction language matched, while answer labels and choice order were preserved across conditions.

If a model provided an explanation, its final answer was extracted when clearly identifiable. Responses with a missing, invalid, or ambiguous A–D selection were marked incorrect. Because commercial interfaces and hidden system prompts may change, exact replication over time may be limited.

3.5 Scoring and Evaluation Metrics

Model outputs were scored by exact match against the human-verified answer key. A response was correct only when its selected answer letter matched the gold answer. When a response included an explanation, we extracted the final answer if it was clearly identifiable. Missing, ambiguous, conflicting, or non-A–D answers were marked incorrect.

For model m and language condition l, accuracy is

$$A_{m,l} = \frac{C_{m,l}}{N} \times 100, \tag{1}$$

where $C_{m,l}$ is the number of correct responses

and $N = 200$. We report Wilson 95% confidence intervals for each language-specific accuracy estimate (Wilson, 1927). Because every item has four answer choices, uniform random selection gives a nominal chance baseline of 25%. We also compute an empirical majority-answer-position baseline that always predicts the most frequent gold-answer position:

$$B_{\text{mode}} = \frac{\max(n_A, n_B, n_C, n_D)}{N} \times 100, \quad (2)$$

where $n_A, \ldots, n_D$ are the frequencies of the four gold-answer positions. This baseline accounts for possible answer-position imbalance.

We additionally report average Arabic accuracy,

$$A_m^{\text{Ar}} = \frac{A_{m,\text{OA}} + A_{m,\text{SA}} + A_{m,\text{EA}}}{3}, \quad (3)$$

the English–Arabic gap,

$$G_m = A_{m,\text{EN}} - A_m^{\text{Ar}}, \quad (4)$$

and maximum cross-language variation,

$$V_m = \max_{l \in L} A_{m,l} - \min_{l \in L} A_{m,l}, \quad (5)$$

where $L = \{\text{EN}, \text{OA}, \text{SA}, \text{EA}\}$. Lower values of G_m and V_m indicate greater cross-lingual final-answer stability. These measures complement language-specific accuracy but do not assess the validity or cross-language equivalence of the models' underlying reasoning processes.

3.6 Experimental Scope and Reproducibility

This study evaluates final-answer accuracy on translated MBE-style multiple-choice questions. It does not evaluate open-ended explanations, citation quality, legal advice, reasoning-chain faithfulness, different legal systems, or professional legal judgment. The findings are therefore limited to the evaluated questions, linguistic conditions, prompts, and model interfaces.

All models were accessed through their official user-facing interfaces in May 2026 using default settings and one run per model–item–language combination. Because commercial models, hidden system prompts, and interfaces may change, the results represent behavior during the recorded evaluation period rather than deterministic fixed-checkpoint performance. Repeated-run stability and temporal variation remain directions for future work.

4 Results

We evaluate model accuracy and robustness across English (EN), Original Arabic/MSA (OA), Simple Arabic (SA), and Egyptian Arabic (EA). Table 2 reports accuracy, average Arabic accuracy, English–Arabic gaps, and cross-language variation. Figure 2 compares English and Arabic performance across models and language conditions.

Because each condition contains 200 questions, we report Wilson 95% confidence intervals alongside the language-specific accuracies in Table 2. Gemini 2.5 Pro performs below the nominal 25% chance baseline in all three Arabic conditions, while Grok performs below chance in Simple and Egyptian Arabic and near chance in MSA. These comparisons describe observed accuracy relative to the nominal baseline and should not be interpreted as formal significance tests. As shown in Figure 2, all evaluated models perform better in English than their average performance across the three Arabic conditions. The English–Arabic gaps remain substantial, reaching 61.2 percentage points for Gemini 2.5 Pro and 23.0 percentage points for Grok.

4.1 English Accuracy Does Not Transfer Reliably to Arabic

All four models achieve their highest accuracy in English, even though the same 200 legal problems, answer choices, legal issues, and correct answers are preserved across all language conditions. This result directly supports the central claim of the paper: LLM legal reasoning is not fully language-invariant.

Qwen obtains the highest English accuracy at 97.0%, but its performance drops to 79.0% in all three Arabic conditions, producing an English–Arabic gap of 18.0 points. GPT-5 is more stable: it scores 90.0% in English and remains between 80.0% and 82.0% across Arabic variants, with the smallest English–Arabic gap of 9.0 points. In contrast, Gemini 2.5 Pro drops from 80.5% in English to 19.3% average Arabic accuracy, producing the largest gap at 61.2 points. Grok also shows a substantial gap, falling from 45.0% in English to 22.0% average Arabic accuracy.

These results show that English performance alone is not a reliable estimate of Arabic legal reasoning performance. A model may appear strong on English legal questions while becoming substantially less reliable when the same questions are

Model	EN	OA	SA	EA	Avg. Ar.	EN–Ar Gap	Max Var.
GPT-5	90.0 (85.1–93.4)	82.0 (76.1–86.7)	81.0 (75.0–85.8)	80.0 (73.9–85.0)	81.0	9.0	10.0
Gemini 2.5 Pro	80.5 (74.5–85.4)	19.5† (14.6–25.5)	21.0† (15.9–27.2)	17.5† (12.9–23.4)	19.3	61.2	63.0
Grok	45.0 (38.3–51.9)	26.5 (20.9–33.0)	16.5† (12.0–22.3)	23.0† (17.7–29.3)	22.0	23.0	28.5
Qwen	97.0 (93.6–98.6)	79.0 (72.8–84.1)	79.0 (72.8–84.1)	79.0 (72.8–84.1)	79.0	18.0	18.0

Table 2: Accuracy (%) and cross-language robustness. Parentheses show Wilson 95% confidence intervals for language-specific accuracies ($N = 200$). † Observed accuracy below the nominal 25% uniform-choice baseline.

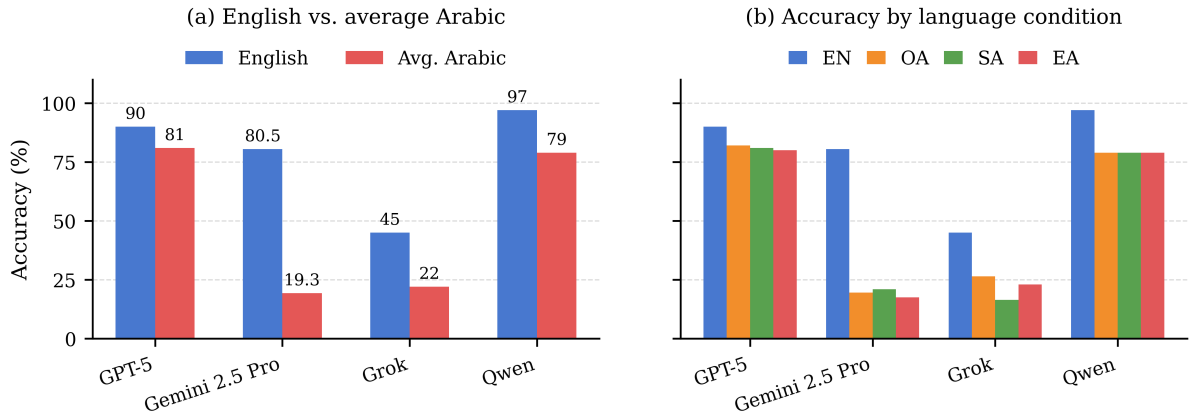


Figure 2: Cross-lingual accuracy across models. (a) English versus average Arabic accuracy. (b) Accuracy disaggregated by language condition: English (EN), Original Arabic (OA), Simple Arabic (SA), and Egyptian Arabic (EA).

expressed in Arabic.

4.2 Accuracy and Robustness Give Different Rankings

The results also show that accuracy and robustness are not the same. Qwen is the strongest model by peak English accuracy and average Arabic accuracy, but GPT-5 is the most robust model across language conditions. GPT-5 has the lowest maximum cross-language variation at 10.0 points, compared with 18.0 points for Qwen, 28.5 points for Grok, and 63.0 points for Gemini 2.5 Pro.

This distinction matters for multilingual legal AI. A model with the highest English score may still show a meaningful English–Arabic gap. Conversely, a model with slightly lower peak accuracy may be more dependable across language varieties. In this benchmark, Qwen is the strongest model by English accuracy, while GPT-5 is the strongest model by cross-language stability.

4.3 Arabic Variants Reveal Model-Specific Weaknesses

The three Arabic conditions reveal different model-specific robustness patterns. GPT-5 shows the most stable Arabic behavior, with only a two-point range across Original Arabic/MSA, Simple Arabic, and Egyptian Arabic. This suggests that GPT-5 is affected by the shift from English to Arabic, but is less sensitive to differences among Arabic registers.

Qwen produces identical accuracy across the three Arabic variants, with 79.0% in Original Arabic/MSA, Simple Arabic, and Egyptian Arabic. This pattern suggests a general English–Arabic transfer gap rather than a weakness tied to one Arabic register or dialect. In other words, Qwen remains strong in Arabic relative to the other models, but its English performance still overestimates its Arabic legal reliability.

Grok is most sensitive to Simple Arabic. Its accuracy drops from 26.5% in Original Arabic/MSA to 16.5% in Simple Arabic, before increasing to 23.0% in Egyptian Arabic. This pattern shows that simplification does not necessarily improve model

Model	Gap	Var.	Pattern
GPT-5	9.0	10.0	Mild sensitivity
Qwen	18.0	18.0	English–Arabic gap
Grok	23.0	28.5	Simple Arabic weakness
Gemini 2.5 Pro	61.2	63.0	Broad Arabic degradation

Table 3: Dominant robustness patterns by model. Gap = English–Arabic gap; Var. = maximum cross-language variation.

reasoning. Instead, simplified wording may change how the model interprets the relationship among facts, rules, and answer choices.

Gemini 2.5 Pro performs poorly across all Arabic conditions, with scores below 22.0%. This indicates broad Arabic degradation rather than a dialect-specific failure. These findings show that Arabic should not be treated as a single evaluation condition. Formal Arabic, simplified Arabic, and dialectal Arabic expose different weaknesses, and evaluating only one Arabic variety may hide important reliability problems.

4.4 Robustness Pattern Analysis

Table 3 summarizes the dominant robustness pattern for each model. The robustness patterns reinforce the main findings. GPT-5 shows only mild cross-lingual sensitivity, while Qwen shows a consistent English–Arabic transfer gap. Grok shows the clearest simplification-sensitive weakness because Simple Arabic is its lowest condition. Gemini 2.5 Pro shows broad Arabic degradation, with low accuracy across all Arabic variants. Overall, the results show that changing the language, register, or dialect can substantially change model performance even when the legal meaning remains fixed. Therefore, multilingual legal evaluation should report not only accuracy, but also English–Arabic gaps and cross-language variation.

4.5 Qualitative Error Inspection

To contextualize the aggregate results, we conducted a preliminary qualitative inspection of Arabic failure cases. This inspection was illustrative rather than exhaustive and focused on cases where the English version was answered correctly but at least one Arabic version was answered incorrectly.

The inspection showed that Arabic errors were not limited to output-format problems. In many cases, models selected a different legal conclusion in Arabic even when the facts, answer choices, and intended correct answer were preserved through legal and linguistic review.

Common failure patterns included confusion between legally similar answer choices, weaker handling of translated legal conditions or exceptions, and instability when the same legal relationship was expressed in simplified or dialectal Arabic. These observations support the quantitative finding that cross-language performance loss reflects robustness limitations rather than only formatting noise.

5 Discussion

5.1 English Accuracy Overestimates Arabic Legal Reliability

All models achieve their highest accuracy in English despite receiving semantically aligned questions across all language conditions. This indicates that performance depends not only on the legal task, but also on its language, register, and dialect. Similar cross-lingual gaps have been reported in multilingual evaluations (Hu et al., 2020; Liang et al., 2020; Bandarkar et al., 2024) and Arabic-focused benchmarks (Abdelali et al., 2024; Koto et al., 2024).

English-only testing may therefore overestimate reliability for Arabic-speaking users. This is especially concerning in legal contexts, where differences in interpreting facts, rules, exceptions, or answer choices can change the legal conclusion (Guha et al., 2023; Katz et al., 2024).

5.2 Accuracy and Robustness Are Different Evaluation Dimensions

Qwen achieves the highest English and average Arabic accuracy, whereas GPT-5 exhibits the smallest cross-language variation. Thus, the most accurate model is not necessarily the most stable across languages. Existing legal benchmarks emphasize task accuracy and legal knowledge, while multilingual benchmarks emphasize cross-language transfer (Guha et al., 2023; Niklaus et al., 2024; Joshi et al., 2024). Multilingual legal evaluations should combine both perspectives by reporting language-specific accuracy, English–Arabic gaps, and cross-language variation.

5.3 Arabic Variety Matters

Models behave differently across Original Arabic/MSA, Simple Arabic, and Egyptian Arabic. GPT-5 and Qwen remain relatively stable, Grok

performs worst in Simple Arabic, and Gemini 2.5 Pro performs poorly across all Arabic conditions.

Formal Arabic evaluation alone may therefore hide important weaknesses. Dialectal Arabic remains underrepresented in LLM evaluation (Mousi et al., 2025), with persistent challenges in dialect identification, generation, and translation (Mousi et al., 2025). Broader Arabic benchmarks also demonstrate the importance of evaluating diverse tasks and linguistic conditions (Abdelali et al., 2024; Koto et al., 2024).

Arabic legal benchmarks should consequently include multiple language varieties. However, our findings are limited to MSA, Simple Arabic, and Egyptian Arabic and should not be generalized to untested dialects.

5.4 Implications for Multilingual Legal AI

Multilingual legal evaluation should test whether models preserve performance across semantically equivalent language versions. Existing benchmarks evaluate legal knowledge, argument reasoning, and rule application (Guha et al., 2023; Abu Shairah et al., 2025), while multilingual legal resources extend evaluation across languages, domains, and jurisdictions (Niklaus et al., 2024; Joshi et al., 2024). However, limited work directly tests whether models preserve their answers across aligned language varieties. Our controlled design shows that model performance changes with language, register, and dialect even when the underlying legal content remains fixed. These results concern final-answer robustness and do not establish the validity or cross-language equivalence of the models' reasoning processes. Such instability may create unequal reliability for users asking equivalent questions in different language varieties. Prior work therefore emphasizes the need for linguistically inclusive evaluation (Hu et al., 2020; Bandarkar et al., 2024; Mousi et al., 2025). Legal AI systems should be evaluated for both accuracy and cross-lingual robustness before use in multilingual, high-stakes settings.

6 Limitations and Future Work

This study evaluates 200 MBE-style multiple-choice questions, four general-purpose models, four aligned language conditions, and one run per model–item–language combination. The findings are therefore specific to this benchmark and should not be generalized to other jurisdictions, Arabic legal systems, legal tasks, language varieties, or models. Because commercial models and their interfaces may change, the results represent an interface-level snapshot from May 2026. Moreover, final-answer accuracy does not demonstrate that models use faithful or equivalent legal reasoning across languages.

Because NCBE materials are protected, we cannot release the original questions, translations, answer choices, or official explanations. Our GitHub repository will instead provide aggregate and anonymized item-level results, model predictions, correctness labels, prompts, scoring and analysis code, and author-created synthetic examples: `https://github.com/hala00001/Rationality-of-Large-language-models-`. These resources support partial, but not exact, reproducibility.

Future work should expand the evaluation to additional legal domains, jurisdictions, task formats, models, and Arabic varieties, including Gulf, Levantine, Iraqi, Moroccan, and Sudanese Arabic. Repeated runs could measure response stability, answer-flip rates, and temporal variation. Future studies should also examine explanations, citations, legal-rule application, and cross-language reasoning consistency, as well as mitigation methods such as Arabic legal retrieval, dialect-aware prompting, self-consistency, and reliability calibration.

7 Conclusion

We evaluated four LLMs on 200 aligned legal questions across English, Original Arabic, Simple Arabic, and Egyptian Arabic. All models performed best in English, but their Arabic performance varied substantially. GPT-5 showed the greatest cross-lingual stability, while Qwen achieved the highest average accuracy. These results demonstrate that strong English performance does not guarantee reliable Arabic legal reasoning, highlighting the need for multilingual and dialect-sensitive evaluation in high-stakes legal AI.

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